IN SILICO EVALUATION OF POLYPODIACEAE METABOLITES AS POTENTIAL INHIBITORS OF CHIKUNGUNYA VIRUS NSP2 PROTEASE

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CERTIFICATE

This is to certify that the Project work entitled "Computational evaluation of Polypodiaceae Metabolites as potential inhibitors of Chikungunya virus nsP2 protease" has been carried out by Ms. Shagun Choudhary, under my supervision and guidance, is pursuing a Master of Science in Botany. I consider that the project has reached the standards and fulfils the requirements of the rules and regulations relating to the nature of the degree. The contents embodied in the thesis have not been submitted for the award of any other degree or diploma in this or any other university.

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1. INTRODUCTION

Chikungunya fever (CHIKF) is a debilitating viral illness spread through the bites of infected mosquitoes. The disease is caused by an alphavirus from the Togaviridae family, which is known for its potential to cause significant discomfort. This virus features an enveloped structure and possesses positive-stranded RNA. The primary carriers of this virus are the *Aedes aegypti* and *Aedes albopictus* mosquitoes, notorious for thriving in tropical and subtropical climates, particularly in urban areas. As a result, outbreaks of Chikungunya fever can lead to widespread illness characterized by severe joint pain, fever, and rashes in affected individuals. The term 'chikungunya' is derived from the Kimakonde word ‘kungunyala’ which means "to become contorted" or "to bend." This reference highlights the characteristic bent posture many patients adopt due to the severe joint pain and debilitating symptoms caused by the Chikungunya virus. First identified in Tanzania in the 1950s, Chikungunya is transmitted primarily through the bites of infected Aedes mosquitoes, mainly Aedes aegypti and Aedes albopictus.

The illness is marked by a sudden onset of fever, severe joint pain, muscle pain, headache, nausea, fatigue, and rash, often leading to long-lasting joint pain that can persist for months or even years. Understanding the origin of the term provides greater insight into the physical impact this disease has on those affected (Schmidt et al, 2022). Chikungunya brings with it a range of rheumatologic symptoms that can lead to a distinctive stooped posture. This characteristic change is a striking reminder of how the disease affects the body, showcasing the challenges faced by those who endure its effects. In the Congo region, the condition is known as "Buka Buka", a term that translates to "broken-broken." This evocative phrase captures the profound and debilitating nature of the joint pain experienced by those afflicted, symbolizing the way it can disrupt daily life and movement, leaving individuals feeling as though their bodies are fragmented and immobilized (Tamfum et al., 2003). The virus is known to cause an acute febrile illness characterized by a sudden onset of fever, accompanied by a distinctive rash and joint pain (arthralgia). After the initial phase, some patients may experience chronic and debilitating arthritic symptoms, which can persist for months or even years. This prolonged condition can significantly impact daily activities and quality of life, making early detection and intervention crucial for managing symptoms and improving overall outcomes (Jafet et al., 2023).

The Chikungunya virus was initially identified during significant outbreaks that emerged in West Africa, where it first captured the attention of public health officials. Chikungunya outbreaks have been notably documented in India during two significant periods: from 1963 to 1973 and again from 2005 to 2019. Some experts suggest that the absence of vigilant surveillance between 1973 and 2005 may have contributed to the dearth of reported Chikungunya virus (CHIKV) cases during those years, highlighting the crucial role that monitoring plays in understanding and managing infectious diseases (Pavri, 1986; Ravi, 2006). The first documented outbreak of the Chikungunya virus (CHIKV) in India occurred in the bustling city of Kolkata in 1963. Following this initial outbreak, the virus spread to other regions, with significant cases reported in Chennai, Puducherry, and Vellore in 1964. These outbreaks marked the beginning of a public health challenge in India, as the virus quickly gained prominence due to its rapid transmission and the associated symptoms, which include high fever, severe joint pain, and rash. The impact of these early outbreaks highlighted the need for improved surveillance and response strategies to control the spread of mosquito-borne diseases in the region (Myers et al, 1965). The 2005–2006 Chikungunya fever (CHIKF) outbreak marked a significant public health concern in India, with 13 states reporting cases during this initial wave. By 2009, the spread of the infection had expanded to 15 states, indicating a troubling trend in the virus's transmission. The situation worsened in subsequent years, as the number of states reporting CHIKF increased to 23 by 2015. This escalation continued, with 28 states affected in 2016, and ultimately reaching 30 states and Union territories by 2019. Notably, the highest number of laboratory-confirmed cases was recorded in 2016, which stood out as a peak year, followed closely by 2017 and 2019, revealing a pattern of resurgence. The data highlights Karnataka, Delhi, and Maharashtra as the states with the most significant burden of confirmed cases, emphasizing the need for enhanced surveillance and control measures in these regions to combat the spread of the disease effectively. These figures underline the ongoing challenges posed by Chikungunya and the importance of public health initiatives to mitigate its impact (TRC report, 2021).

The genome of the Chikungunya virus (CHIKV), which belongs to the RNA alphavirus family, is a linear, positive-sense, and single-stranded RNA molecule that measures approximately 11.8 kilobases (kb) in length. This genome encodes several essential proteins responsible for viral replication and assembly, including the non-structural proteins that facilitate viral RNA synthesis and the structural proteins that form the viral capsid (Powers et al., 2010). The genetic structure of CHIKV consists of two distinct open reading frames (ORFs). The first ORF is

responsible for encoding four non-structural proteins: nsP1, nsP2, nsP3, and nsP4, each playing a crucial role in viral replication and the formation of the viral replication complex. Specifically, nsP1 is involved in RNA capping, nsP2 has helicase activity, nsP3 is a multifunctional protein that can modulate host immune responses, and nsP4 serves as the RNA-dependent RNA polymerase. The second ORF encodes five structural proteins that are essential for the viral architecture and infection process. This includes the capsid protein (C), which encapsulates the viral RNA and is integral to the formation of the viral particle. Additionally, it produces the peptide 6K/TF, which has a role in the assembly and release of the virus.

The three envelope proteins—E1, E2, and E3—contribute to the formation of the viral envelope, facilitate viral entry into host cells, and are key targets for neutralizing antibodies. Together, these proteins orchestrate the lifecycle of the virus, from assembly to infection, highlighting the complexity and intricacy of viral machinery (Konishi, 1980; Simizu, 1984). The nsP123 precursor, composed of non-structural proteins nsP1, nsP2, and nsP3, along with nsP4, collaborates in the synthesis of a full-length negative-strand RNA intermediate during the viral replication cycle. Once this negative-strand RNA is synthesized, nsP123 undergoes a series of precise proteolytic cleavages, which sequentially release its constituent proteins. This processing is crucial for the subsequent transcription of positive-strand genomic RNA (49S) and sub-genomic RNA, which are essential for viral replication and the production of viral proteins. This intricate interplay between the proteins ensures efficient replication and the establishment of a successful viral life cycle, allowing the virus to effectively hijack the host's cellular machinery (Fros et al., 2010). nsP1 plays a critical role in the viral life cycle by exhibiting RNA capping characteristics, which are essential for the stabilization and translation of viral RNA. This protein is instrumental in facilitating the synthesis of the negative strand of the viral RNA, thereby contributing to the overall replication process of the virus.

In contrast, nsP2 is pivotal in shutting down host cell transcription, which is crucial for evading the host's immune response. This protein achieves its function through its multifaceted activities, including RNA helicase, RNA triphosphatase, and protease activities, all of which dismantle the host's cellular machinery. Additionally, nsP4 serves as the viral RNA polymerase, orchestrating the replication of the viral genome. Meanwhile, nsP3 is an integral component of the replicase complex, collaborating with other non-structural proteins to ensure effective viral replication and transcription. Together, these non-structural proteins form a sophisticated machinery that undermines host defense while promoting viral propagation. (Strauss et al., 1994). The Chikungunya virus (CHIKV) non-structural protein 2 (nsP2) is crucial for the viral

life cycle and contributes significantly to the process of viral infection. In cells that have been infected with CHIKV, it has been observed that the majority of the nsP2 proteins, which are integral to the replication complex (RC), are not concentrated in the cell membranes where early viral replication occurs. Instead, these proteins are predominantly distributed throughout the cytoplasm and the nucleus of the host cells. This widespread localization suggests that nsP2 may have multiple roles in facilitating viral replication and manipulating host cell machinery, potentially influencing processes such as RNA synthesis and evasion of the host's immune responses (Frolova et al., 2010). The non-structural protein 2 (nsP2) in the nucleus disrupts antiviral signaling by hijacking the host's transcriptional machinery, impairing normal gene expression, and compromising immune responses. This interference not only allows the virus to evade detection but also induces cytopathic effects in vertebrate cells, leading to cellular damage and promoting viral replication (Fros et al., 2010; Akhrymuk, 2012; Frolov, 2018). As the largest protein encoded by the alphavirus genome, nsP2 plays a critical role in viral replication and pathogenesis. Specifically, the nsP2 protein of the Chikungunya virus (CHIKV) exhibits dual functionalities, possessing both 5′–3′ RNA helicase activity, which is essential for unwinding nucleic acids during replication, and RNA annealing activities that facilitate the stabilization of intermediate RNA structures (Das et al., 2014). Additionally, the C-terminal region of nsP2, which spans amino acids 471 to 798, contains distinct enzymatic domains, including a protease responsible for cleaving viral polyprotein precursors and a methyltransferase-like (MTL) domain that is thought to play a role in modifying RNA structures for enhanced stability and translation efficiency. These multifunctional properties make nsP2 a pivotal component in the CHIKV life cycle and a potential target for antiviral therapies. (Rausalu et al., 2016). Despite the successful determination of the structures of the C-terminal protease region of alphavirus nsP2 (designated as nsP2p) by several studies (Russo et al., 2006; Narwal et al., 2018), and the uncleaved P23 precursor protein, which lacks the N-terminal helicase domain of the nsP2 (referred to as nsP2h) (Shin et al., 2012), the three-dimensional structure of the chikungunya virus (CHIKV) nsP2h remains to be elucidated. The absence of this structural information hinders our understanding of the functional mechanisms and interactions of this crucial protein in the viral life cycle and pathogenesis. Advancing the study of CHIKV nsP2h is essential for the development of targeted antiviral strategies and therapeutic interventions.

Chikungunya virus (CHIKV) poses a significant global public health threat, primarily transmitted by two species of mosquitoes: *Aedes aegypti* and *Aedes albopictus*. Mosquitoes of

the *Aedes aegypti* species are particularly prevalent in tropical and subtropical regions globally, thriving in urban environments characterized by human activity and stagnation of water sources. As a critical vector, *Ae. aegypti* is known to transmit several significant arboviruses, including the dengue virus (DENV), Zika virus, and yellow fever virus, making it a major public health concern (Soni et al., 2023). This species typically resides in urban areas, where it adapts to indoor environments and primarily feeds on humans, contributing to its effectiveness in spreading diseases (Soper, 1967). Notably, *Ae. aegypti* is largely confined to warmer climates, as it lacks the physiological adaptations necessary to survive winter conditions and cannot overwinter in temperate zones (Reinhold et al., 2018; Hawley et al., 1989). This limitation emphasizes the importance of climatological factors in the population dynamics and geographical distribution of this mosquito species, as well as its role in the epidemiology of mosquito-borne diseases in affected regions. In contrast, the Asian Tiger mosquito, scientifically known as *Aedes albopictus*, is a notable vector for various arboviruses, including dengue fever, chikungunya, and Zika virus. This species is capable of thriving in a wide range of environments, as it inhabits not only subtropical regions but also extends its presence into cooler, temperate areas. Remarkably, *Aedes albopictus* has developed adaptive strategies that allow it to survive in colder climates by overwintering in both egg and adult stages.

During winter months, the female mosquitoes can lay fertilized eggs in water sources that can freeze, while adults may enter a state of dormancy in sheltered locations, ensuring their survival until favorable conditions return in the spring. This resilience significantly contributes to their geographic expansion and raises concerns about disease transmission in diverse ecosystems (Hawley et al., 1989). As such, *Ae. albopictus* has a larger geographic range and tends to live outdoors and further from humans, feeding on other animals in nature. Interestingly, CHIKV has also been detected in other *Aedes species* such as *Ae. furcifer*, *Ae. taylori*, *Ae. luteocephalus*, *Ae. africanus*, *Ae. dalzieli* and *Ae. neoafricanus* and also in *Anopheles coustani*, *Anopheles rufipes,* and *Culex ethiopicus (*Diallo et al., 1999*).* Vector competence, the ability of an arbovirus vector to acquire a pathogen and successfully transmit it to another susceptible host, is a complex process influenced by external factors including temperature, the availability of vertebrate hosts, vector population density, and predation, as well as internal factors including mosquito survival and virus replication. For CHIKV, horizontal transmission through saliva that is injected when a mosquito probes or feeds on blood is the most common transmission mechanism, although vertical transmission via infected eggs may also occur at a low rate (Agarwal et al., 2014).

Chikungunya virus (CHIKV) is transmitted through two distinct ecological cycles: the urban cycle and the sylvatic cycle. The urban cycle involves the transmission of the virus between humans and mosquitoes, where an infected human transmits the virus to the mosquito through bites. In turn, the mosquito can then bite and infect other humans, creating a cycle predominantly linked to urban environments, where high human density increases the risk of transmission. Conversely, the sylvatic cycle occurs in forested or rural areas and involves wildlife, especially primates and other forest-dwelling animals. In this cycle, an infected animal transmits CHIKV to mosquitoes, which can subsequently bite humans, leading to outbreaks. (Singh et al., 2011). Chikungunya virus (CHIKV) has a well-documented route of infection within the human body, characterized by a preference for specific cell types that exhibit heightened susceptibility to viral entry and replication. Notably, human epithelial cells-found in tissues that line organs and structures-and endothelial cells-forming the inner lining of blood vessels-are particularly conducive to CHIKV infection. Additionally, primary fibroblasts, which play a crucial role in tissue maintenance and repair, have shown significant vulnerability to the virus. Furthermore, monocyte-derived macrophages, which are immune cells that engulf pathogens and debris, are capable of supporting CHIKV replication. In contrast, certain immune cells, including lymphoid and monocytoid cells, alongside primary lymphocytes and monocytes, demonstrate resistance to CHIKV replication and do not facilitate the virus's life cycle. Similarly, monocyte-derived dendritic cells, which are pivotal in antigen presentation and the activation of the adaptive immune response, have not been shown to replicate the virus (Sourisseau et al., 2007).

The immune response to chikungunya virus (CHIKV) infection has been partially elucidated. Upon infection, CHIKV initiates an immune response by indirectly stimulating the production of type I interferon (IFN), a crucial antiviral cytokine. This process primarily involves the activation of non-hematopoietic cells, notably fibroblasts, which play a vital role in the body's defense against the virus. The activation of these fibroblasts leads to the secretion of type I IFNs, which in turn facilitate the recruitment and activation of various immune cells, essential for effectively clearing CHIKV from the systemic circulation. While interferon regulatory factors (IRFs) 3 and 7 functioned redundantly in adult immune responses, research has shown that neonates with deficiencies in either of these factors exhibited a markedly increased susceptibility to viral infections. This vulnerability is particularly concerning given the critical role IRFs play in the production of interferons and the subsequent activation of innate immune pathways. In infants, the impaired signaling through these pathways suggests that their immune

systems are less equipped to effectively respond to pathogenic challenges, highlighting the importance of both IRF3 and IRF7 in early-life immunity (Schilte et al., 2012). Chikungunya virus (CHIKV) seems to initiate a complex signaling cascade by activating interferon promoter stimulator 1 (IPS-1), a key mediator in the innate immune response. This activation results in the accumulation of interferon regulatory factor 3 (IRF3)-dependent messenger RNAs (mRNAs), which are vital for the expression of type I interferons and other antiviral proteins. However, intriguingly, CHIKV also appears to inhibit the translation of these mRNAs into functional proteins. This dual action may allow the virus to evade host immune responses, fostering its replication while simultaneously triggering an immune signaling response (White et al., 2011).

CD8+ lymphocytes, also known as cytotoxic T cells, are predominantly present in the skin rashes observed in patients during the acute phase of their condition. These cells play a crucial role in directly targeting and destroying infected or altered cells. In contrast, CD4+ T-cells, which are primarily helper T cells, constitute the majority of the immune cell population found in the synovial effusions of patients with chronic conditions. These CD4+ T-cells are essential for orchestrating the immune response, facilitating communication between immune cells, and promoting inflammation and tissue repair within the joints (Pardigon, 2009). BST-2, also known as bone marrow stromal antigen 2, is a significant protein that may play a protective role in lymphoid tissues during viral infections. Research suggests that BST-2 is involved in modulating the host's inflammatory responses triggered by Chikungunya virus (CHIKV) infection. By regulating these inflammatory responses, BST-2 could help maintain the integrity of the lymphatic system and potentially mitigate the severity of symptoms associated with CHIKV, thereby influencing the overall immune response and disease progression. (Fernandez et al., 2014).

Research indicates that inflammation resulting from chikungunya virus (CHIKV) infection can significantly impact the proliferation and functionality of bone-forming cells known as osteoblasts, as well as bone-resorbing cells called osteoclasts. This disruption may play a significant role in the chronic effects observed in individuals with long-term CHIKV infection. Cytokines such as tumor necrosis factor-alpha (TNF-α), interleukin-6 (IL-6), and interleukin-1 (IL-1) are key inflammatory mediators released during the infection. These cytokines not only augment osteoclast activity—thereby promoting bone resorption—but are also critically involved in the process of osteoclastogenesis, the formation of new osteoclasts (Chen et al., 2015).

Diagnosis can often be delayed because the symptoms can closely resemble those of dengue fever or Zika virus infection. In particular, the presentation of fever accompanied by polyarthralgia (widespread joint pain) demonstrates noteworthy clinical significance, yielding an 84% sensitivity in identifying cases. This means that 84% of individuals with the condition will present these symptoms. Furthermore, this symptom combination carries a 71% positive predictive value (PPV), indicating that when these symptoms are present, there is a 71% chance that the diagnosis is correct. Conversely, it also has an 83% negative predictive value (NPV), implying that 83% of individuals without these symptoms are indeed not suffering from the condition. These statistics underscore the importance of careful clinical evaluation to avoid misdiagnosis and ensure timely and appropriate management (Thiberville et al., 2013). Enzyme-linked immunosorbent assays (ELISA) are highly effective diagnostic tools utilized to confirm the presence of anti-CHIKV (Chikungunya virus) antibodies in patient samples. Typically, immunoglobulin M (IgM) antibody levels peak between three to five weeks following infection, serving as a crucial indicator of recent exposure to the virus. These IgM antibodies can remain detectable for up to two months, which is essential for clinical assessments and understanding the timeline of infection. In addition to serological testing, polymerase chain reaction (PCR) techniques can be employed to identify and genotype the virus directly from clinical specimens. This molecular method plays a significant role in tracking viral strains and understanding transmission patterns, enhancing our ability to respond effectively to outbreaks of Chikungunya virus infections.

Current treatment approaches primarily aim to alleviate the severity of symptoms associated with the disease, rather than providing a definitive cure. The management of symptoms typically relies on the administration of antipyretics, such as acetaminophen or ibuprofen, which help to reduce fever and discomfort. In addition to these medications, non-steroidal anti-inflammatory drugs (NSAIDs) play a crucial role in decreasing inflammation and pain, thereby improving the patient's overall quality of life. This symptom-focused strategy allows patients to manage their experiences while ongoing research seeks more effective therapies that could eventually lead to a cure (Rougeron et al., 2015). Corticosteroids for treating acute conditions have shown mixed results, prompting caution among clinicians. While these medications can provide significant anti-inflammatory effects and alleviate symptoms initially, there is considerable concern regarding potential rebound effects, where symptoms may worsen after the treatment is discontinued. This hesitancy is compounded by the risk of side effects associated with corticosteroid use, which may include increased infection risk, weight gain,

and hormonal imbalances. As a result, healthcare providers often weigh the potential benefits against the possible long-term impacts before incorporating corticosteroids into a treatment regimen (Goupil et al., 2016). In the management of severe chronic arthralgia, a range of disease-modifying anti-rheumatic drugs (DMARDs) have been proposed to alleviate symptoms and slow disease progression. Among these, methotrexate is commonly used due to its ability to modulate the immune response and decrease inflammation. Hydroxychloroquine is another DMARD that is particularly effective in cases related to autoimmune conditions, as it can help with both pain relief and the prevention of disease flare-ups. Sulphasalazine, often utilized for its anti-inflammatory properties, is beneficial for patients with inflammatory arthritis. Each of these medications may have varying effects depending on the individual patient's condition and response, and they are often prescribed alongside non-steroidal anti-inflammatory drugs (NSAIDs) for enhanced pain management. Close monitoring by healthcare professionals is essential to tailor the treatment plan and mitigate any potential side effects (Bouquillard et al., 2009).

Pteridophytes, commonly known as ferns and their allies, represent a significant group of vascular plants that occupy a unique evolutionary position between non-tracheophytes (such as mosses and liverworts) and spermatophytes (seed-producing plants). This diverse group encompasses over 48 families and approximately 587 genera, showcasing various forms and habitats. Pteridophytes are predominantly found in moist, shaded environments, where they thrive in forest understories and along riverbanks. They play essential ecological roles, contributing to biodiversity and soil stabilization, and as indicators of environmental health. Worldwide, there are estimated to be over 12000 species of pteridophytes, highlighting their ecological significance and the need for conservation efforts to protect their habitats (Christenhuz et al., 2016; Zaman et al., 2019).

Most species are predominantly found in tropical regions, characterized by warm temperatures and high humidity, which provide ideal conditions for biodiversity. These regions are often home to dense rainforests, vibrant coral reefs, and lush ecosystems that support a vast array of plant and animal life. Following closely are moist temperate regions, where a moderate climate and sufficient rainfall create a diverse range of habitats, such as deciduous forests and wetlands. Subtropical regions, with their warm summers and mild winters, also contribute significantly to global biodiversity, harboring unique species adapted to these distinct environments (Khullar, 2000).

Vegetables, traditional remedies, and landscaping are some of the primary uses of these plants. They are commonly cultivated in home gardens and commercial farms for their nutritional value, offering a rich source of vitamins and minerals. Additionally, many varieties have been utilized in folk medicine for centuries, providing natural treatments for a range of ailments. In landscaping, they are prized not only for their aesthetic appeal but also for their ability to attract beneficial wildlife, improve soil health, and enhance the overall biodiversity of gardens and green spaces (Ranil et al., 2021). Edible pteridophytes, commonly known as ferns, are not only a unique culinary ingredient but also a rich source of essential nutrients. They contain significant amounts of proteins, which are vital for muscle repair and growth, as well as various vitamins that support overall health, including vitamins A and C, which are important for immune function and skin health, respectively. Additionally, these ferns provide crude fibre, which aids in digestion and promotes gut health.

Furthermore, edible pteridophytes are enriched with an array of minerals such as calcium, iron, and magnesium, all of which play crucial roles in bone health, energy production, and muscle function. Beyond their nutritional benefits, these plants also harbor numerous bioactive compounds, including steroids and terpenoids, which have been studied for their potential anti-inflammatory and antioxidant properties. They contain phenolic acids, known for their ability to combat oxidative stress, and flavonoids, which may support cardiovascular health and possess anti-cancer properties. Overall, the diverse composition of edible pteridophytes makes them a valuable addition to a balanced diet (Chettri et al., 2018).

Polypodiaceae, commonly known as the polypody family, is recognized as one of the richest fern families, with 65 genera and 1652 species distributed across continents except Antarctica, boasting a remarkable diversity that contributes significantly to the rich tapestry of tropical and subtropical forests. This family encompasses a wide variety of genera, including well-known groups such as *Polypodium*, *Pyrrosia*, and *Lepisorus*, showcasing an impressive range of morphological forms and ecological adaptations. Members of Polypodiaceae thrive in various habitats, from shaded understories to open, sunlit areas, often establishing themselves as epiphytes on trees or as ground-dwelling ferns. Their abundance and adaptability play a crucial role in the overall biodiversity of these ecosystems, contributing to soil stabilization, humidity regulation, and providing microhabitats for various organisms (Schneider et al., 2004). During the Cenozoic radiation period, which began around 66 million years ago following the mass extinction event that wiped out the dinosaurs, leptosporangiate ferns experienced significant diversification within an ecosystem increasingly dominated by flowering plants (angiosperms).

This dynamic interaction allowed these ferns to exploit various niches within the rich, layered canopy that characterized the new botanical landscape. As a result, the Polypodiaceae family, known for its wide range of forms and ecological adaptations, emerged as a diverse lineage within this advantageous environment, showcasing an impressive array of species that thrived in the humid, shaded conditions of tropical and subtropical forests. This period marked a critical phase in the evolutionary history of ferns, highlighting their adaptability and resilience in response to changing climatic and ecological conditions (Schuettpelz and Pryer, 2009).

The majority of species in the *Polypodiaceae* family are epiphytes, meaning they grow on other plants, typically trees without drawing nutrients from them. This family is renowned for its remarkable diversity, encompassing a wide range of forms and adaptations. *Polypodiaceae* represents one of the most extensive and abundant groups of pantropical vascular epiphytes, thriving in tropical and subtropical forests. These ferns play a crucial role in their ecosystems, often contributing to biodiversity by providing habitats for various organisms and participating in nutrient cycling within the forest canopy. Their adaptability to different light conditions and moisture levels allows them to flourish in the rich, humid environments found in these lush forests (Gentry and Dodson, 1987; Benzing, 2004).

Despite the significant biological activities attributed to the *Polypodiaceae* family reported in various studies, such as antidiabetic (*Phymatopteris triloba* (Houtt.) Pic. Serm.) (Chai et al., 2013), anticancer (*Polypodium leucotomos*) (Gonzalez et al., 2010) and anti-inflammatory activities (*Polypodium leucotomos*) (Choudhry et al., 2014) there are many other *Polypodiaceae* ferns not yet characterized or studied specifically for their potential medical uses. This is the case with *Dryopteris dickinsii*, *Dryopteris crassirhizoma*, *Dryopteris filix-mas*, *Pteridium aquilinum*, *Pteris multifida*, *Pteris cretica*, *Pteris dactylina*, and *Pteris semipinnata*.

Secondary metabolites are small organic molecules that emerge from primary metabolites during the developmental stages of a plant, typically characterized by molecular masses of less than 3000 daltons (da). These compounds play crucial roles in plant physiology, ecology, and evolution. The chemical nature and composition of these metabolites can significantly differ among plant species, influenced by factors such as genetic makeup, environmental conditions, and developmental stages. Defining the boundary between primary and secondary metabolites can be challenging, as there is considerable overlap, and many compounds found in natural plant products are classified as secondary metabolites. Unlike primary metabolites, which are

essential for basic cellular functions and are directly involved in growth and development (e.g., sugars, amino acids, and nucleotides), secondary metabolites often serve specialized functions. These include roles in defense against herbivores and pathogens, attraction of pollinators, and competition with other plant species for resources. Secondary metabolites are of great interest for several reasons most notably for their structural diversity and potential as drug candidates and antioxidants. They encompass a wide range of chemical classes, including alkaloids, flavonoids, terpenes, and phenolics, each exhibiting unique biological activities. Notably, many secondary metabolites possess potent anti-inflammatory, antibacterial, antiviral, and anticancer properties, making them valuable in pharmaceutical research and development. However, the complexity of plant metabolites poses significant challenges for industrial synthesis. Many of these compounds are difficult, if not impossible, to replicate in a laboratory setting due to their intricate structures and specific biosynthetic pathways that are often adapted to the unique ecological niches of the plants that produce them (Sanchez et al., 2000; Bourgaud et al., 2001).

Secondary metabolites in plants are classified into several distinct chemical groups, primarily based on their biosynthetic pathways. These groups include phenolic compounds, terpenes and steroids, and nitrogenous compounds. The unique chemical structures of these metabolites play a crucial role in their functions, as they are involved in various biological processes, including defense mechanisms, signaling, and interactions with other organisms. Phenolic compounds, for instance, are known for their antioxidant properties and their role in protecting plants from UV radiation and herbivory. Terpenes and steroids contribute to the plant's aromatic qualities and can deter herbivores or attract pollinators. Meanwhile, nitrogenous compounds, such as alkaloids and amino acids, often have toxic effects on herbivores and can serve as a means of chemical defense. Plants face a multitude of environmental stresses, such as drought, pathogenesis, herbicide exposure, salinity, and heavy metal pollution. These stresses trigger signaling pathways that promote the accumulation of secondary metabolites as a form of adaptation and resilience. For example, during drought conditions, certain phenolic compounds may increase to enhance the plant's water retention capabilities. Similarly, when exposed to pathogens or herbicides, the synthesis of specific alkaloids may be upregulated to provide a chemical defense. Overall, the production of secondary metabolites is a vital strategy for plants to survive and thrive in challenging environments (Eid et al., 2015).

There are three major classes of plant metabolites based on the biosynthetic pathway: (i) phenolic groups (composed of simple sugars and benzene rings), (ii) terpenes and steroids

(composed mainly of carbon and hydrogen), and (iii) nitrogen-containing compounds (Saxena et al., 2013).

Phenolic compounds, which originate from the phenylpropanoid pathway in plants, exhibit a wide range of structures and play a crucial role in providing color to flowers, fruits, and vegetables. These compounds contribute not only to the aesthetic appeal of plants but also to their ecological interactions, such as attracting pollinators.

Alkaloids and flavonoids represent two significant groups of secondary metabolites that are characterized by their remarkable structural diversity and numerous biological functions. Alkaloids, often derived from amino acids, are known for their pharmacological effects and can serve as natural defense mechanisms for plants against herbivores and pathogens. Flavonoids, on the other hand, are involved in various plant processes, including UV protection, signaling, and pigmentation. A wide variety of phenolic metabolites have been identified to possess several beneficial properties, including potent anti-oxidant, anti-carcinogenic, and anti-inflammatory effects. These biological activities suggest that phenolic compounds may contribute to health benefits for humans, making them valuable components of a diet rich in fruits and vegetables. Research continues to explore their potential therapeutic applications and mechanisms of action, highlighting the importance of these secondary metabolites in both plant biology and human health (Huang et al., 2009).

Alkaloids represent a significant class of secondary metabolites that are renowned not only for their pharmacological properties but also for their recreational uses. These nitrogen-containing compounds often exhibit diverse biological activities, influencing various physiological processes in humans and other organisms. In addition to alkaloids, another important group of nitrogen-based organic compounds is closely related to aromatic amino acids, which are synthesized via the shikimic acid pathway. This metabolic route is vital for the production of several key biomolecules. Among these aromatic amino acids, phenylalanine and tryptophan serve as fundamental precursors, playing crucial roles in the biosynthesis of both phenolic compounds and alkaloids. Phenylalanine is a precursor to many important substances, including tyrosine, which is critical for synthesizing neurotransmitters like dopamine, norepinephrine, and epinephrine. Tryptophan, on the other hand, is essential for the production of serotonin, a neurotransmitter involved in mood regulation. The interconnectedness of these metabolic pathways underscores the complexity of plant secondary metabolism and its significance in health and disease (Sanchez et al., 2000).

Flavonoids, a diverse group of polyphenolic compounds, encompass over 4,000 different types that have been identified from plant sources. These compounds are particularly abundant in various green plants (Mabry et al., 2012). Flavonoids predominantly exist in the form of glycosides, which means they are bound to sugar molecules, and are primarily found in the leaves, flowers, stems, and roots of plants. Structurally, flavonoids are characterized by the presence of two benzene rings, contributing to their complex chemical properties. Major classes of flavonoids include chalcones, flavones, flavanols, flavanones, anthocyanins, and isoflavones, many of which exhibit vibrant colors that are often responsible for the hues of fruits and flowers. For instance, anthocyanins provide red, purple, or blue pigments, while flavanones are often found in citrus fruits and contribute to their tangy flavors (Harborne, 1998).

Phenylpropanoids are a diverse group of chemical compounds primarily synthesized by plants and certain microorganisms through the shikimate pathway. This metabolic route is crucial as it leads to the production of essential aromatic amino acids, particularly phenylalanine and tyrosine. These amino acids play a vital role in various biological processes, including protein synthesis and the production of neurotransmitters.

Since mammals, including humans, cannot synthesize these amino acids de novo, they must acquire them through their diet. Consequently, plants serve as a fundamental component of the food chain, providing these essential building blocks for life. The presence of phenylpropanoids also contributes to the flavor, fragrance, and overall nutritional quality of the plants, further enhancing their importance in the human diet. Additionally, some phenylpropanoids possess antioxidant properties, which can offer health benefits to those who consume plant-based foods rich in these compounds (Hasan et al., 2022).

Terpenes are a structurally diverse and extensive group of metabolites, with over 35,000 distinct terpenes characterized to date (Dewick, 2002). These compounds are primarily composed of isoprene units—five-carbon building blocks—linked together in various configurations. The structural versatility of terpenes arises from their ability to undergo cyclization reactions, leading to the formation of cyclic compounds that are easily identifiable. Terpenes are classified into different categories based on the number of isoprene units present in their carbon skeleton, including monoterpenes (two isoprene units), sesquiterpenes (three isoprene units), diterpenes (four isoprene units), and so on (Cseke et al., 2006). This classification reflects the complexity and diversity of terpenoid structures, which can

significantly affect their biological activity. In addition to their structural diversity, terpenoids have been found to exhibit a wide range of biological activities, making them of particular interest in pharmacology and medicinal chemistry. Some terpenoids demonstrate antitubercular properties, providing potential therapeutic avenues against tuberculosis (Garcia et al., 2006). Others have shown promising anticancer effects, indicating their potential role in cancer treatment strategies (Dewick, 2002). Moreover, certain terpenes exhibit anxiolytic effects, potentially aiding in the management of anxiety disorders, while some have been identified as mutagenic, warranting further investigation into their safety profiles and environmental impact (Medina et al., 1997).

Alkaloids are a diverse group of nitrogen-containing organic compounds that typically exhibit a cyclic structure. Despite their significant pharmacological importance, their occurrence in nature is relatively limited (Pelletier, 1983). These compounds are predominantly soluble in aqueous solutions, which facilitates their extraction through processes that involve the protonation of their nitrogen atoms. This property makes them particularly amenable to extraction techniques used in both research and pharmaceutical applications. Among the most well-known alkaloids are caffeine, nicotine, cocaine, and morphine, each recognized for their distinct physiological effects on the central nervous system. Caffeine, the psychoactive compound found in coffee and tea, is known for its stimulant properties, while nicotine, primarily derived from tobacco, acts as both a stimulant and a relaxant. Cocaine, a potent stimulant, has profound effects on mood and energy but is also associated with high addictive potential, and morphine, an effective analgesic, is widely used for pain management but has a significant risk of dependence and abuse (Dewick, 2002). Despite representing a small fraction of the vast array of plant metabolites, alkaloids account for approximately 50% of all plant-derived pharmaceuticals available on the market (Oksman et al., 2004).

The utilization of plant metabolites dates back to around 2600 BC, marking the beginning of a long history where these compounds played significant roles in human society. Over the following 4000 years, secondary metabolites derived from plants were primarily employed for medicinal purposes, as poisons, and as food additives or flavorings. Among these natural products, morphine, which is extracted from the opium poppy (Papaver somniferum), was the first secondary metabolite to be isolated in 1806. This groundbreaking discovery not only revolutionized the field of pharmacology by introducing powerful analgesics but also paved the way for extensive research into the vast variety of plant-derived compounds, sparking a

new era in the exploration of secondary metabolites and their potential benefits and risks (Sanchez et al., 2000).

According to the ancient texts of Sushruta and Charka, which form the foundation of Ayurvedic medicine, certain ferns are specifically prescribed for their medicinal properties. These references can be found in the Samhita texts, where various ferns are identified for their therapeutic uses in the treatment of ailments. In addition, Unani medicine also recognizes the value of pteridophytes, utilizing these ferns for their health benefits. Both traditional systems highlight the importance of these plants in promoting well-being and address various health conditions through the application of their unique properties (Uddin MG et al., 1998). Inhibition of the cyclooxygenase (COX) pathway is an important mechanism through which certain fern secondary metabolites, particularly flavonoids, exhibit their anti-inflammatory properties. These flavonoids are a class of polyphenolic compounds widely found in various ferns, and they play a crucial role in plant defense. By inhibiting the COX enzymes, which are key mediators in the inflammatory response, these compounds reduce the production of prostaglandins—molecules that contribute to inflammation and pain. This suggests that the therapeutic potential of fern-derived flavonoids lies in their ability to modulate inflammatory pathways, offering insights for the development of natural anti-inflammatory agents in medicine (Liang et al., 1999).

Pteridophytes, a group of vascular plants that includes ferns, horsetails, and clubmosses, possess a remarkable array of bioactive compounds that contribute to their medicinal properties. Research has demonstrated that these plants exhibit antioxidative characteristics, which help neutralize free radicals in the body, thereby protecting cells from oxidative stress and reducing the risk of various chronic diseases. Additionally, certain species of Pteridophytes have shown promise in cancer prevention, with studies indicating that their phytochemicals may inhibit the growth and proliferation of cancer cells. Furthermore, these plants also possess antibacterial activities, making them valuable in the development of natural antimicrobial agents to combat infections. Overall, the therapeutic potential of Pteridophytes highlights their significance in traditional medicine and their promise for future pharmaceutical applications (Amaral et al., 2009; Govindappa et al., 2011; Lin et al., 2008; Lopez, 2009; Yoshida et al., 2008).

2. REVIEW OF LITERATURE

Chikungunya fever is a mosquito-borne viral illness caused by the Chikungunya virus (CHIKV), which primarily spreads through the bite of infected Aedes mosquitoes, mainly Aedes aegypti and Aedes albopictus. A sudden onset of high fever characterizes the disease, often reaching up to 39°C (102°F), accompanied by severe joint pain, muscle pain, fatigue, rash, and headache. The joint pains can be debilitating and may persist for weeks or even months after the initial infection. The virus was first identified in the 1950s during an outbreak in southern Tanzania, where it was linked to a significant public health crisis. Since then, Chikungunya has re-emerged and expanded to various parts of the world, including Africa, Asia, Europe, and the Americas, often resulting in massive epidemics that affect large populations. Notably, environmental changes, increased urbanization, and global travel have contributed to the spread of the virus, making it a growing concern for public health authorities. Efforts to control the disease involve vector management and public health awareness campaigns to reduce mosquito breeding and prevent infections. (Silva et al., 2018).

In 1953, after the isolation of Chikungunya virus (CHIKV) from the sera of infected patients, as well as from the mosquito species Aedes (Stegomyia) aegypti and various Culex species, researchers classified this virus into the arbovirus group A (Ross,1956; Calisher and Karabatsos, 1998). Chikungunya virus (CHIKV) is currently classified within the Alphavirus genus, which is part of the Togaviridae family. It is recognized as a member of the Semliki Forest Virus (SFV) antigenic complex, a group of viruses that share similar structural and genetic characteristics. CHIKV is primarily known for its transmission via mosquito vectors, mainly Aedes aegypti and Aedes albopictus, leading to significant outbreaks in tropical and subtropical regions. The virus is notable for causing high fever and severe joint pain in infected individuals. Understanding the taxonomy and classification of CHIKV within the SFV complex helps in the development of targeted vaccines and therapeutic strategies (Richter et al., 2015; ICTV, 2017).

Shortly after its identification in East Africa in 1952, Chikungunya virus (CHIKV) was subsequently documented in several regions of Africa, including central Africa, specifically in Uganda, and further south in the sub-Saharan region (Weinbren et al., 1958). The development of nucleic acid sequencing technologies and molecular evolution tools has facilitated the classification of CHIKV strains based on their geographical origin. These strains are

categorized into three primary groups: the East, Central, and South African lineage (ECSA) (Powers et al., 2000).

Phylogenetic analysis of Chikungunya Virus (CHIKV) isolated during outbreaks from 1958 to 1973 across various regions in Asia revealed that these strains clustered into a distinct monophyletic group, which has been categorized as the Asian lineage (Powers et al., 2000; Volk et al., 2010). In 1963, India experienced its first documented outbreak of a significant viral infection in Kolkata, where researchers successfully isolated the virus for the first time (Jadhav et al., 1965; Kumar and Gopal, 2010).

Throughout the last millennium, one notable epidemic emerged prominently in the early 1960s, wreaking havoc across various regions in India. In 1963, Kolkata faced a significant outbreak that alarmed health authorities and overwhelmed local medical facilities. This was followed by a similar outbreak in 1965, which spread to multiple cities, including Pondicherry and Chennai in Tamil Nadu, where the densely populated urban areas contributed to the rapid transmission of the disease. The epidemic further extended its reach to the coastal town of Rajahmundry and the bustling cities of Visakhapatnam and Kakinada in Andhra Pradesh. The impact was also felt inland, with Sagar in Madhya Pradesh reporting cases, while Nagpur, a major city in Maharashtra, grappled with the repercussions of this widespread health crisis. Each of these locations experienced unique challenges in controlling the outbreak, highlighting the variations in healthcare infrastructure, public health responses, and the socio-economic factors that influenced the epidemic's trajectory across the region (Schuffenecker et al., 2006).

In the early 1970s, a notable outbreak of Chikungunya occurred, specifically in 1971, and sporadic cases continued to emerge across various regions of Maharashtra until 1973. This initial surge in cases marked a significant episode in the history of the disease. Following the last documented outbreak in 1973, there was a notable decline in reported cases, leading to a period of inactivity where no outbreaks were recorded until 2005, suggesting a long hiatus from the disease's impact on public health. However, in 2006, after nearly three decades without significant activity, the situation dramatically shifted, as India reported approximately 13.9 million suspected cases of Chikungunya, alongside 2,001 laboratory-confirmed cases. The first outbreak in the early 1970s was attributed to the Asian genotype of the Chikungunya virus (CHIKV), while the resurgence in 2005–2006 was identified as being caused by the East/Central/South African (ECSA) genotype (Kumar and Gopal, 2010; Sarkar et al., 1964; Shah et al., 1964; Kalantri et al., 2006; Sudeep and Parashar, 2008).

The World Health Organization (WHO) recently confirmed the resurgence of the fever in India, noting that its attack rate varies significantly, ranging from 4% to 45% across different regions. This resurgence could be attributed to several extrinsic factors, such as changes in climate that influence the breeding patterns of disease vectors, increased human mobility, and varying public health responses. Additionally, socio-economic conditions, including lack of access to healthcare and sanitation, have further compounded the situation, making certain populations more vulnerable to outbreaks (Kumar and Gopal, 2010; Kalantri et al., 2006).

Numerous factors have been proposed to explain the re-emergence of the virus. One significant contributor is the E1-A226V mutation in the viral envelope, which has enhanced the virus's adaptability to Aedes albopictus mosquitoes, also known as tiger mosquitoes. This mutation poses a particular risk to vulnerable populations such as the very young and the elderly, whose immune systems are either underdeveloped or compromised due to age-related factors. The absence of herd immunity facilitated a rapid and widespread transmission of the virus across various states in India, highlighting the challenges faced in controlling outbreaks. Following an initial surge in cases, the incidence gradually declined in 2014; however, trends indicated that susceptibility to the virus persisted, with cases continuing to fluctuate until 2018 (TRC, 2021).

Between 2005 and 2008, India experienced a significant outbreak of Chikungunya, with thirteen states reporting cases of this mosquito-borne viral disease. The impact of the outbreak was considerable, leading to widespread public health challenges and prompting increased surveillance and vector control efforts across the affected regions. Fast forward to 2018, the situation escalated further, as reports indicated that thirty-two states had documented cases of Chikungunya, highlighting its pervasive presence throughout the country.

The latest data on the incidence and confirmed cases of Chikungunya have been provided by the National Center for Vector-Borne Diseases Control (NCVBDC) for the period between 2018 and 2023. This ongoing monitoring is crucial for understanding the trends of the disease, guiding prevention strategies, and informing healthcare responses to mitigate the impact of future outbreaks. Between 2005 and 2008, India experienced a significant outbreak of Chikungunya, with thirteen states reporting cases of this mosquito-borne viral disease. The impact of the outbreak was considerable, leading to widespread public health challenges and prompting increased surveillance and vector control efforts across the affected regions. Fast

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In 2018, the state of Karnataka reported the highest number of suspected chikungunya cases, totaling 20411. Following closely in second place was Gujarat, with 10601 suspected cases, while Maharashtra recorded 9,884 suspected cases, placing it third. Despite the high number of suspected cases, the confirmed cases in Karnataka were relatively low at 2,546. In Gujarat, only 1,290 cases were confirmed, and Maharashtra had 1,009 confirmed cases. Notably, this indicates a significant discrepancy between suspected and confirmed cases. Only 12.4% of suspected cases in Karnataka were confirmed, with Gujarat exhibiting a slightly lower confirmation rate of 12.1%, and Maharashtra having the lowest confirmation rate at 10.2%. The following year, 2019, saw a dramatic increase in suspected chikungunya cases, with the total reaching 81,914; however, only 12,205 of these cases were confirmed across the country. Karnataka again led the counts, with 43,698 suspected cases, yet only 3,664 were confirmed, demonstrating a continuing trend of low confirmation rates amidst high suspicion. In contrast, several states reported alarmingly low confirmed case numbers. Assam, Haryana, and Mizoram recorded zero confirmed cases, and other states like Manipur, Uttarakhand, and Jammu and Kashmir reported only a handful of confirmed cases: 3, 1, and 0, respectively. D&N Haveli and Chandigarh also had no confirmed cases, emphasizing a troubling gap in the understanding and management of this disease across various regions.

In 2020, the total number of suspected Chikungunya cases in India experienced a significant decline, dropping from 81,914 cases in the previous year to just 43,424 cases, marking a reduction of approximately 53%. Among the states, Karnataka reported the highest number of suspected cases at 16,111, followed by Gujarat with 8,120 cases and Maharashtra with 4,258 cases. Notably, Karnataka, Gujarat, and Rajasthan also recorded the highest number of confirmed cases, indicating a persistent trend in the geographical spread of the disease. However, in 2021, there was a dramatic resurgence in suspected cases of Chikungunya, with reports indicating an increase by 2.7 times compared to the previous year. This spike raised concerns among health authorities and highlighted the ongoing challenges in controlling

vector-borne diseases. Throughout this period, Gujarat, Karnataka, and Maharashtra consistently remained at the forefront, experiencing both the highest numbers of suspected and confirmed cases, underscoring the need for enhanced surveillance and preventive measures in these regions.

In 2022, the number of suspected chikungunya cases saw a marked increase, while the number of confirmed cases experienced a notable decline. The states of Gujrat, Karnataka, and Maharashtra consistently reported high numbers of suspected cases throughout the year. Notably, Puducherry also emerged as a significant hotspot, recording over 10,000 suspected cases. In contrast, both Himachal Pradesh and Lakshadweep had not documented any chikungunya cases before 2022. However, this year marked a shift, as Himachal Pradesh reported a total of 253 suspected cases, with only 3 confirmed, whereas Lakshadweep reported 103 suspected cases with no confirmations. Transitioning into 2023, there was a significant decrease in both suspected and confirmed chikungunya cases across the country. Remarkably, Nagaland, which had previously been classified as a non-endemic state, reported its first cases of chikungunya, indicating that the virus had spread nationwide. Karnataka reported the highest number of confirmed cases this year, but overall, the public health data indicated a decline in the prevalence of chikungunya compared to the previous year, reflecting improved awareness and preventive measures being implemented across various states (NCVBDC, 2023).

Chikungunya virus (CHIKV) is characterized as an enveloped virus with a diameter ranging from 60 to 70 nanometers, featuring an icosahedral nucleocapsid structure. The viral genome is composed of a positive-sense single-stranded RNA (ssRNA) molecule, which is approximately 11.8 kilobases in length. This genome is organized into two primary open reading frames (ORFs), designated ORF1 and ORF2, which are crucial for encoding viral proteins. The 5′ end of the genome is capped, an essential modification that aids in translation and stability, while the 3′ untranslated region (3′ UTR) terminates with a polyadenylated tail, further contributing to mRNA stability and regulation during the viral life cycle. ORF1 encodes polyprotein precursors that subsequently generate four nonstructural proteins: nsP1, nsP2, nsP3, and nsP4. These nonstructural proteins are pivotal for viral replication and host interaction. In contrast, ORF2 encodes a set of six structural proteins, including the capsid protein (C) and envelope proteins E1, E2, E3, and 6K, as well as the transframe proteins (TM) (Kendall et al., 2019; Khan et al, 2002).

The chikungunya virus genome is encapsulated within virions that exhibit structural similarities to those found in other members of the alphavirus family. While the specific cellular receptors responsible for chikungunya virus entry remain unidentified, research indicates that chikungunya virions are primarily internalized through clathrin-mediated endocytosis—a process involving the formation of vesicles that transport molecules into the cell. Additionally, existing evidence suggests that the entry mechanism may vary depending on the cell type, indicating the possibility of alternative pathways being utilized for viral entry (Duijl et al., 2015).

Chikungunya virus RNA translation occurs before replication, resulting in the formation of non-structural (ns) polyproteins P123 and P1234. The nsP2 region's protease activity converts the nonstructural polyproteins into mature nonstructural proteins (Utt et al., 2015). Like other alphaviruses, mature nsP1 contributes to viral replication. Moreover, it possesses enzymatic properties, including RNA triphosphatase and NTPase. In addition to having enzymatic activities, including NTPase and RNA triphosphatase activity (Karpe et al., 2011), RNA helicase activity (Das et al., 2014), and protease activity (Utt et al., 2015), nsP2 can counteract host cellular antiviral responses through multiple mechanisms. These mechanisms include general transcriptional shutdown by degradation of the Rpb1 subunit of host cellular RNA polymerase II (Akhrymuk et al., 2012), as well as more specific mechanisms including interference with JAK/STAT signalling (Fros et al., 2010; Fros et al., 2013), the unfolded protein response (Fros et al., 2015), and autophagy (Judith et al., 2013).

The three-dimensional structure of the N-terminal (macro) domain of nsP3 has been resolved and shown to bind negatively charged polymers, including RNA (Malet et al., 2009). Increasing evidence also suggests that the intrinsically unstructured C-terminal region of nsP3 acts as a binding platform for numerous host cellular proteins, including G3BP proteins24,28 and amphiphysins (Neuvonen et al., 2011), and that these interactions are important for virus infection. Despite the absence of direct evidence, nsP4 of chikungunya virus clearly appears to be an RNA-dependent RNA polymerase.

Considerable attention has been dedicated to the identification of host proteins that interact with the non-structural proteins of chikungunya virus (Bourai et al., 2012; Treffers et al., 2015; Das et al., 2014; Rathore et al., 2014). Cellular proteins also interact with viral RNA (Dickson et al., 2012). In most cases, the functional importance of these interactions remains unknown. However, some interactions have a positive effect on virus replication, whereas others mediate

antiviral effects. Some interacting proteins might have both proviral and antiviral effects. For example, the interaction of nsP3 of alphaviruses with G3BP proteins has an antiviral effect by preventing the formation of stress granules (Panas et al., 2012). A proviral function for G3BP proteins has been shown, with depletion of these proteins resulting in inhibition of early replication (Scholte et al., 2015).

One possible role of G3BP proteins is to facilitate switching from non-structural protein translation to replication of genomic RNA (Scholte et al., 2015). However, the exact details of RNA replication of chikungunya virus have not been studied. By analogy with Semliki Forest virus and Sindbis virus, replication first generates negative-strand RNA, probably existing exclusively in duplex form with the positive-strand genome, and then generates numerous positive—genomic and subgenomic RNAs (Rupp et al., 2015). Genomic RNA contains a packaging signal that, unlike most of the alphaviruses, is located in the region encoding nsP2 (Kim et al., 2011).

Structural proteins, which are essential for virion formation, are translated from subgenomic RNA. Chikungunya infection shuts down translation of cellular mRNAs, and subgenomic RNA remains the only actively translated mRNA in the cell (Scholte et al., 2013). Unlike Semliki Forest virus and Sindbis virus, the subgenomic RNA of chikungunya virus does not have a stable stem-loop structure, the so-called capsid enhancer, at its 5′ region (Merits A, unpublished). Thus, how active translation of chikungunya virus RNA is maintained remains unclear. The first structural protein, the capsid protein, is not only responsible for nucleocapsid formation but also possesses nuclear export and import signals, allowing entry to and exit from the nucleus (Thomas et al., 2013). The glycoprotein part of the structural polyprotein is translated by membrane-associated ribosomes and is processed and post-translationally modified by cellular enzymes. The assembly and budding of chikungunya virions takes place on the plasma membrane of infected vertebrate cells. This process is sensitive to the antiviral effects of the cellular protein tetherin (Ooi et al., 2015), which are counteracted by nsP1 (Jones et al., 2013). In polarised cells, the release of chikungunya virions occurs at the apical domain of cells; data obtained with different inhibitors suggest that the N-glycans of chikungunya virus envelope glycoproteins could serve as apical sorting signals (Lim et al., 2014).

As the emerging disease Chikungunya virus (CHIKV) is causing serious global health implications due to its socioeconomic impact and high morbidity rates, treatment for CHIKV remains complex (Kumar et al., 2020; Constant et al., 2021; Silva et al., 2018; Zeller et al.,

2016; CDCP, 2022; Sales et al., 2018; WHO, 2022). CHIKV is an arbovirus in the *Togaviridae* family and is characterized by the National Institute of Allergy and Infectious Diseases as a category B priority pathogen, which is the second highest priority of biological agents (CDCP, 2022; WHO, 2022; NIAID, 2022). Currently, treatment for CHIKV is focused on relieving symptoms and is complex as there is no current dedicated vaccine or treatment (Kumar et al., 2020; Burt et al., 2014).

Acute infection has a 2-7 day incubation period, and approximately 95% of infected individuals develop symptoms, which typically resolve within about two weeks (CDCP, 2022; WHO, 2022; Goupil, 2016; Ali and Combe, 2011). During the acute stage, arthralgia is similar to that caused by dengue or Zika virus, so serological detection is necessary for a definitive diagnosis before treatment can be administered (Silva et al., 2018; WHO, 2022). CHIKV rarely leads to death, and when it does, it usually only occurs in the most vulnerable population of elderly individuals with comorbidities and children (Castro and Lima, 2016; CDCP, 2022).

The demand for treatment and vaccine research for CHIKV remains elevated as the high rate of morbidity caused by the virus creates significant psychosocial and economic impacts, which affect the health and well-being of individuals, populations, and the globe as a whole (Constant et al., 2021; Silva, 2018). There is an urgent need for the development of alternative natural and pharmaceutical countermeasures to this emerging viral infection that can be used in low-resource areas of the world where this virus affects the most (Kumar et al., 2020; Puntasecca et al., 2021).

Integrated control of epidemiological surveillance and environmental management to eliminate mosquito breeding is an important factor in reducing the transmission of CHIKV (Silva et al., 2018). Preventive measures are the most effective protection for environmental management of mosquito vectors and include wearing long sleeves to minimize the amount of skin surface exposed to mosquito bites, using mosquito nets, elimination of mosquito breeding sites like standing water where mosquitoes could lay eggs, use of insecticides and the installation of window and door screens in homes (Kumar et al., 2020; CDCP, 2022; WHO, 2022; Goupil and Mores, 2016; Ali and Combe, 2011). However, total prevention of transmission from environmental management is impossible due to CHIKV misdiagnoses and underdiagnoses in many regions and its disproportionate effects in low-resource areas (Kumar et al., 2020; Silva et al., 2018).

When preventative measures fail, a person's acute CHIKV symptoms are generally treated with fluids, rest, and pharmaceutical medications for pain management determined by a physician. Antipyretics and analgesics are pharmaceutical drugs that have many contraindications and not everyone can use them for pain management, and they have many side effects that can occur from chronic use (CDCP, 2022; WHO, 2022).

Currently, there is no single recommended treatment therapy for CHIKV (Amaral et al., 2020). As a result, many drugs have been used with varying success. Many focus on supportive treatment and reducing joint pain (Handy et al., 2022). Some of the most used pharmaceutical therapies for CHIKV include non-steroidal anti-inflammatory drugs (NSAIDS), disease-modifying antirheumatic drugs (DMARDs), and antivirals.

Over-the-counter NSAIDs are often used to relieve mild arthralgia pain and reduce fever in CHIKV-positive patients and are currently the most routinely used treatment (Burt et al., 2017). A majority of patients are reactive to NSAIDS, with about 89% of patients experiencing some immediate pain relief (Rosario et al., 2015). Although many physicians rely on NSAIDs as a means to control symptoms, they may provide limited pain relief and are recommended only in mild cases, usually a one to three on the visual analog scale (VAS) for pain (Cunha and Trinta, 2017). There is no conclusive evidence that moderate NSAID treatment in more severe cases shows any efficacy (Goupil and Mores, 2016).

Another drug used for CHIKV is MTX (Methotrexate). MTX serves as the foundation of most rheumatic arthritis (RA) treatments due to its anti-inflammatory properties (Parashar and Cherian, 2014; Pandya, 2008; Foissac et al., 2015). These properties come from inhibition of proinflammatory cytokines interleukin (IL) 1, IL-6, IL-8, and tumor necrosis factor alpha (TNFa), which are correlated with disease severity in both RA and CHIKV-related arthralgia (Amaral et al., 2020; Ravindran and Alias, 2017). MTX can be prescribed to patients for months or even years post-CHIKV infection (Bedoui et al., 2018).

Ribavirin is an antiviral that is being researched for therapeutic benefits in CHIKV patients (Sales et al., 2018). With ribavirin as their treatment, patients with polyarthritis from a chikungunya infection reported improvement, claiming a reduction in pain and joint swelling. Patients expressed an increase in progress with walking (70% in patients vs 30% in the placebo group) along with a reduction of edema (80% vs 60%) (Ravichandran and Manian, 2008). When compared to ribavirin, 6-azauridine has proven to be more effective against CHIKV even

when used at low concentrations (Parashar and Cherian, 2014; Briolant et al., 2004; Rada and Dragun, 1977).

Pteridophytes are spore-bearing vascular plants. They are diverse and disparate set of plants, today comprising approximately 11,000 extant species of ferns (including horsetails and whisk ferns) and 1300 species of lycophytes (Schuettpelz et al., 2016). The pteridophyte macrofossil record extends back to the late Silurian (Gensel, 2008), and pteridophytes were ecologically dominant for hundreds of millions of years as the first plants to evolve roots or leaves (Schneider et al., 2002), to move into drier habitats, and to form forests (Bateman et al., 1998; Stein et al., 2012).

Pteridophytes are distinguished from other plants by being vascular, dispersing by spores, and by their life cycle: both the sporophyte and gametophyte phases are free-living and nutritionally independent (Taylor et al., 2005; Haufler et al., 2016). Additionally, because they reproduce and disperse without the aid of biotic intermediates, they are less susceptible to biotic factors that could obscure the relationship between their geographic ranges and underlying abiotic conditions. Finally, the high dispersal potential of their spores reduces the effect of historical contingency on taxon ranges.

Their deep evolutionary history and tight connection to abiotic factors make pteridophytes an ideal system for studying evolutionary and ecological processes through time. In addition, they have an excellent macrofossil record in both temporal range and quality. Pteridophytes tend to grow close to depositional environments with greater fossilization potential, and thus their fossil record is denser, at least for the first few hundred million years of land plant evolution, than for groups like the seed plants, where much of the morphological innovation is postulated to have occurred in drier habitats where the chance of fossilization is low (Looy et al., 2014). Also, pteridophytes have an added advantage for paleobotanical studies in that their reproductive organs frequently have a long-persistent association with their leaves and stems, which greatly facilitates whole-plant reconstructions.

Polypodiaceae, often referred to as the polypody family, is acknowledged as one of the most diverse families of ferns, comprising 65 genera and 1,652 species found on every continent except Antarctica. This family exhibits a remarkable variety that significantly enhances the biodiversity of tropical and subtropical forests. It includes notable genera such as Polypodium, Pyrrosia, and Lepisorus, which display a wide array of morphological characteristics and ecological adaptations. Members of the Polypodiaceae family flourish in diverse environments,

ranging from shaded forest understories to bright, open spaces, frequently growing as epiphytes on trees or as terrestrial ferns. Their prevalence and versatility are vital to the overall biodiversity of these ecosystems, aiding in soil stabilization, humidity control, and the creation of microhabitats for numerous organisms (Schneider et al., 2004).

The Polypodiaceae family, commonly known as the sword ferns, occupies a vital niche within the plant kingdom, characterized by its extensive diversity and adaptability. Comprised of over 700 species, this family illustrates remarkable evolutionary traits that enable plants to thrive in a variety of habitats, indicating their significant ecological roles. Notably, the classification of ferns, including those within the Polypodiaceae, has proven challenging, with ongoing discussions regarding familial and generic concepts (Christenhusz et al., 2011). This complexity underscores the importance of a standardized classification system, which can enhance the organization and study of these plants. Furthermore, the Polypodiaceae family contributes to global biodiversity, offering ecological services such as soil stabilization and habitat provision for various organisms, while also holding cultural significance in many regions (Baker et al., 2011). Thus, understanding the Polypodiaceae is crucial not just for botanical sciences, but also for conservation efforts worldwide.

The Polypodiaceae family, one of the most diverse within the fern clade, presents a complicated classification that reflects the evolutionary relationships of ferns as a whole. This family is situated within the larger Pteridophyta division, which encompasses all ferns. The Polypodiaceae is characterized by a wide range of morphological features, from the finely divided fronds of the genus Polypodium to the robust, leathery leaves often seen in species like Microsorum. Understanding these variations allows researchers to explore phylogenetic connections within the Pteridophyte lineage. Recent studies have highlighted the significance of genetic and molecular data in reassessing traditional classifications, emphasizing the need for a revised taxonomy that better represents the complexities observed in fern evolution (Sirimalwatta et al., 2019), (Wang et al., 2014). As a result, the ongoing research and classification efforts are crucial for elucidating the relationships not only within Polypodiaceae but across the entire fern family tree.

The Polypodiaceae family, renowned for its rich diversity, is marked by several key morphological features that distinctly categorize its members. One prominent characteristic is the leaf structure, which varies from simple to highly dissected fronds, often displaying a pinnate or pinnatifid arrangement. These fronds may also exhibit a range of textures, from

leathery to herbaceous, often adapted to their specific habitats. Additionally, the family is recognized for its unique sporangial arrangement; members generally possess sori that are either exposed or covered by indusia, playing a vital role in spore dispersal. Furthermore, the prevalence of epiphytic growth among many species, especially in tropical environments, highlights their adaptations to diverse substrates and ecological niches. The ongoing research into the genus Microsorum and further classification within the Polypodiaceae family underscores the significance of these morphological attributes in understanding evolutionary relationships and biodiversity (Chen and Chuan, 2022; Sirimalwatta et al., 2019).

Secondary metabolites are specialized compounds that contribute to the survival and ecological success of plants (Kajla et al., 2023). These metabolites are characterized by their low molecular weight and are synthesized through a variety of biosynthetic pathways, constituting what is known as secondary metabolism (Seca and Trendafilova, 2021). The study of these compounds has garnered significant attention due to their potential applications in medicine, agriculture, and industry, where they serve as antibiotics, agrochemicals, flavors, fragrances, dyes, pigments, and food preservatives.

3. Aims and Objectives

The main objective of the current study is to screen the secondary metabolites of different species of the Polypodiaceae family as potential inhibitors for Chikungunya Virus (CHIKV) nsP2pro. The following are the major objectives of the current study –

3.1 To find the extant species of the Polypodiaceae family.

3.2 To prepare a metabolite library of secondary metabolites from different Polypodiaceae species.

3.3 To analyse the ADME properties of metabolites.

3.4 To carry out the molecular docking for analysis of interactions among metabolites and nsP2pro.

4. METHODS AND MATERIALS

4.1 SECONDARY METABOLITES LIBRARY

Through a review of the literature, a secondary metabolites library was created in order to identify possible Chikungunya treatments. The final secondary metabolites library contains 546 metabolites. The secondary metabolites library contains PubChem ID, metabolite names, IUPAC names, molecular weight, molecular formula, canonical smiles, and references of these metabolites were retrieved from the PubChem Database.

PubChem (https://pubchem.ncbi.nlm.nih.gov) is a public repository for information on chemical substances and their biological activities. Since its launch in 2004 as a component of the US National Institutes of Health (NIH) 's Molecular Libraries Roadmap Initiatives, PubChem has rapidly grown into a key chemical information resource serving scientific communities in various areas, including cheminformatics, chemical biology, medicinal chemistry, and drug discovery.

PubChem contains one of the largest corpora of publicly available chemical information. As of September 2015, it has more than 157 million depositor-provided chemical substance descriptions, 60 million unique chemical structures, and 1 million biological assay descriptions, covering about 10 thousand unique protein target sequences. PubChem organizes this vast amount of data into three inter-linked databases: Substance, Compound, and BioAssay. The Substance database (https://www.ncbi.nlm.nih.gov/pcsubstance) stores depositor-contributed information. Unique chemical structures are extracted from the Substance database and stored in the Compound database (https://www.ncbi.nlm.nih.gov/pccompound). The BioAssay database (https://www.ncbi.nlm.nih.gov/pcassay) stores descriptions of biological assays on chemical substances. The primary identifiers for the Substance, Compound, and BioAssay databases are SID (SubstanceID), CID (CompoundID), and AID (AssayID), respectively.

PubChem's data are provided by more than 350 contributors (https://pubchem.ncbi.nlm.nih.gov/sources/), including university labs, government agencies, pharmaceutical companies, chemical vendors, publishers, and a number of chemical biology resources. The data provided by these contributors is not just limited to small molecules, but also includes siRNAs, miRNAs, carbohydrates, lipids, peptides, chemically modified macromolecules, and many others.

Although PubChem was initially conceived as a central repository for the NIH's Molecular Libraries Program (MLP), it includes data contributed by numerous non-MLP organizations. For example, PubChem contains a substantial amount of literature-derived bioactivity data of chemical substances, manually extracted from tens of thousands of scientific articles by data contributors such as ChEMBL (Bento et al., 2014) and BindingDB (Liu et al., 2007). In addition, through integration with data from DrugBank (Law et al., 2014), the Hazardous Substances Data Bank (Fonger et al., 2014) and other databases, the annotation of chemical records includes pharmacology, drug target information, toxicology, safety, and handling information.

PubChem's three primary databases (i.e., Substance, Compound and BioAssay) and the data flow between them. Each data contributor provides descriptions on chemical samples using PubChem Upload (see the 'Data submission using PubChem Upload' section). PubChem calls these community-provided chemical sample descriptions 'substances' and stores them in the Substance database (https://www.ncbi.nlm.nih.gov/pcsubstance). Different substance records (provided by the same or different contributors) may contain different kinds of information for the same molecule. For example, one substance record may provide information on biological functions of glucose, while another may describe characteristics of a research grade sample of glucose. The Substance database stores these different descriptions about the same molecule as separate records that are independent of each other. The Substance database maintains the provenance of substance records, helping users see who provided what information to PubChem.

The bioactivity of a chemical sample may be measured using various experimental techniques and conditions. These experimental factors may substantially affect the accuracy and precision of the measured bioactivity of a molecule. Especially, considering that the bioactivity data in PubChem are collected from many different data sources, these data would have a very limited use unless detailed information on experimental protocols is provided together with the data. Therefore, the descriptions of biological experiments on chemical substances are stored in a separate database called the BioAssay database (https://www.ncbi.nlm.nih.gov/pcassay). The depositor-provided information on an assay may have multiple bioactivity outcomes determined with different samples (i.e., different *substances*) of the same molecule. Therefore, PubChem provides a comprehensive overview of the biological activity profile of the tested compounds by generating associations between the test results and tested compounds (via tested substances).

4.2 SWISSADME

This website allows you to compute physiochemical descriptors as well as to predict ADME parameters, pharmacokinetic properties, druglike nature, and medicinal chemistry friendliness of one or multiple small molecules to support drug discovery. In this study, SwissADME was used to predict the ADME properties of the selected compounds.

The SwissADME web tool presented here is freely accessible at http://www.swissadme.ch and is meant for user-friendly submission and easy analysis of the results, also for non-experts in CADD. SwissADME strong points are, non-exhaustively: different input methods, computation for multiple molecules, and the possibility to display, save, and share results per individual molecule or through global, intuitive, and interactive graphs. Finally, SwissADME is integrated into the SwissDrugDesign workspace.

A large variety of *in silico* methods share the objective of predicting ADME parameters from molecular structure (Tian et al., 2015). Noteworthy, the pioneer work of Lipinski *et al*. examined orally active compounds to define physicochemical ranges for high probability to be an oral drug (i.e. the drug-likeness) (Lipinski et al., 2001). This so-called *Rule-of-five* delineated the relationship between pharmacokinetic and physicochemical parameters. Whereas physicochemical parameters give a global description of the structure, molecules can be directly described by substructure searches. These techniques are at the root of Structural Alert (Brenk et al., 2008), the PAINS (Baell et al., 2010), or the Lilly MedChem (Bruns et al., 2012) filters applied to cleanse chemical libraries from compounds most likely unstable, reactive, toxic, or prone to interfere with biological assays because unspecific frequent hitters, dyes or aggregators (Irwin et al., 2015). We used Swiss-ADME to assess the ADME profiles of 447 selected metabolites and then conducted further investigations. Lipinski's rule-compliant compounds were docked with nsP2pro.

4.3 PROTEIN AND LIGAND PREPARATION

OpenBabel was used to generate 3D structures of selected metabolites from canonical SMILES inputs. Open Babel was used to add explicit hydrogen atoms and generate a PDB structure (O'Boyle et al., 2011). Open Babel tools in PyRx were used for energy minimization and optimization of the molecules, and the resulting structures were saved in PDBQT format (Dallakyan and Olson 2015). The RCSB database's 3D coordinates for nsP2pro (PDB ID: 4ZTB) were retrieved (https://doi.org/10.2210/ pdb4ZTB/pdb). Eliminating insignificant molecules like water was performed using UCSF Chimera for further investigation (Pettersen

et al. 2004). Protein optimization was carried out employing Discovery Studio (Dassault Systèmes, USA), and the resulting structures were stored in PDB format. The MGL tool was chosen to add polar hydrogen and Kollman charges to the molecule, and the results were stored in PDBQT file format.

PyRx is a Virtual Screening software for Computational Drug Discovery that can be used to screen libraries of compounds against potential drug targets. PyRx enables Medicinal Chemists to run Virtual Screening from any platform and helps users in every step of this process - from data preparation to job submission and analysis of the results. While it is true that there is no magic button in the drug discovery process, PyRx includes docking wizard with easy-to-use user interface which makes it a valuable tool for Computer-Aided Drug Design. PyRx also includes chemical spreadsheet-like functionality and powerful visualization engine that are essential for Rational Drug Design.

4.4 MOLECULAR DOCKING OF LIGANDS

Molecular docking is a computational procedure that attempts to predict noncovalent binding of macromolecules or, more frequently, of a macromolecule (receptor) and a small molecule (ligand) efficiently, starting with their unbound structures, structures obtained from MD simulations, or homology modeling, etc. The goal is to predict the bound conformations and the binding affinity. To identify new potential inhibitors for the nsp2 pro of Chikungunya virus, the site-specific docking-screening of 272 compounds was done using AutoDock Vina.

Molecular docking for the identification of potential antiviral molecules against nsp2 pro was performed by using AutoDock Vina (Trott and Olson,2010). The grid centre point was set as X = 20.223 Y = 1.321 Z =-16.419. The box dimensions were 16 Å 16 Å 16 Å, and exhaustiveness was set at 16. The other parameters were left at their default values. The binding energy was distributed from -2.1 to -7.2 kcal/mol. The molecules having the least binding energies were analyzed for docking interactions. AutoDock and AutoDock Vina (The Scripps Research Institute) are some of the most widely used, free, open-source tools for molecular docking simulations. AutoDock is a collection of command-line programs that can be employed to predict binding conformations of a small flexible ligand to a macromolecular target whose structure is known. This technique combines the rapid grid-based method used for energy evaluation with conformation searching and simulated annealing.

AutoDock 4 was used for molecular docking previously. The new AutoDock Vina has a more accurate binding algorithm that can speed up the rate by approximately 2 orders of magnitude

as compared to AutoDock 4. In addition, AutoDock 4 has significantly improved predictions of binding mode, assessed by the training tests employed in AutoDock 4. By the use of multithreading on the multicore machines, faster processing can be achieved from parallelism. AutoDock Vina clusters the results for the user transparently and automatically calculates the grid maps.

PyMOL has been widely used for 3D visualization of macromolecules and it becomes one of the most popular tools for preparing high-resolution images of macromolecules for publications. However, many users may ignore or are not aware of many other useful modules and functions, which perfectly facilitate the CADD. For instance, PyMOL supports 12 different stereo visualization modes, which enables the users to conceive much more 3D information for the atom positions in space. This useful function, inspiring new ideas for the compound modification, is seldom used. Instead, many users only stick to the 2D mode, rotating the models with their mouse. The stereo view is of great importance for in situ design. For instance, to obtain detailed information about the binding pocket for the design of novel molecules, the protein surface can be shown as a mesh.

5. RESULT AND DISCUSSION

5.1 Selection and Screening of Secondary Metabolites from *Polypodiaceae*:

The Polypodiaceae family is known for its rich phytochemical composition, including flavonoids, alkaloids, and carotenoids, which exhibit diverse biological activities such as antiviral, anti-inflammatory, and antioxidant effects. Various species of the *Polypodiaceae* family are selected for their secondary metabolite extraction. The secondary metabolite library was successfully prepared from eight Polypod species: *Dryopteris dickinsii, Dryopteris crassirhizoma, Dryopteris filix-mas, Pteridium aquilinum, Pteris multifida, Pteris cretica, Pteris dactylina, and Pteris semipinnata.* Among these, *Pteris semipinnata* exhibited the highest metabolite diversity, containing 206 identified compounds, followed by *Dryopteris dickinsii* with 116 metabolites, *Dryopteris crassirhizoma* with 114, *Pteris cretica* with 52, *Pteris dactylina* with 38, *Pteridium aquilinum* with 13, *Dryopteris filix-mas*, and *Pteris multifida* with 1 metabolite.

5.2 Library Preparation:

A total of 546 Secondary metabolites from *Polypodiaceae* were retrieved from the PubChem database and literature survey and compiled into a metabolite library, which contains Polypodiaceae species with PubChem ID, IUPAC name, molecular weight, molecular formula, scientific name, and canonical smiles of compounds. The chemical structures were converted into PDBQT format for docking.

5.3 ADME analysis:

To determine drug-likeness and pharmacokinetic properties, ADME (Absorption, Distribution, Metabolism, and Excretion) analysis was performed on 447 molecules. Based on Lipinski's Rule of Five, 272 molecules passed the criteria, indicating favorable oral bioavailability.

5.4 Molecular Docking analysis:

The 208 phytomolecules subjected to virtual screening based on ADME parameters showed binding energy ranging from -2.1 to -7.2 kJ/mol. The molecules having the least binding energies (-6.2 kcal/mol and lower) were analysed for docking interactions.

S.No.	Metabolite Name	Binding Energy
1.	3'-Geranylchalconaringenin	-7.2

2.	[(2S)-2-[(2R,3S,4S)-3,4-dihydroxy-5-oxooxolan-2-yl]-2-hydroxyethyl] 3-(4-hydroxyphenyl)prop-2-enoate	-6.7
3.	[(2S)-2-[(2R,3S,4S)-3,4-dihydroxy-5-oxooxolan-2-yl]-2-hydroxyethyl] 3-(4-hydroxy-3-methoxyphenyl)prop-2-enoate	-6.6
4.	Desmethylxanthohumol	-6.5
5.	(2Z)-3-(4-{[(2S,3R,4S,5S,6R)-3,4,5-trihydroxy-6-(hydroxymethyl)oxan-2-yl]oxy}phenyl)prop-2-enoic acid	-6.4
6.	4-(beta-D-glucosyloxy)benzoic acid	-6.3
7.	Resveratrol	-6.3
8.	5'-Prenylxanthohumol	-6.3
9.	1-(5-Hydroxy-7-methoxy-2,2-dimethyl-3,4-dihydrochromen-6-yl)-3-(4-hydroxyphenyl)prop-2-en-1-one	-6.3
10.	ent-9-Hydroxy-15-oxokaur-16-en-19-oic acid beta-D-glucopyranosyl ester	-6.3
11.	Epicatechin	-6.2
12.	Apigenin	-6.2

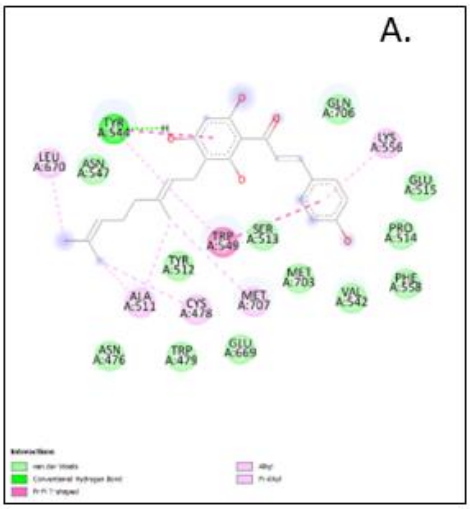
5.5 Interpretation of Docking Results:

The presence of van der Waals interactions enhances the stability of the ligand within the binding pocket, ensuring a proper fit. These interactions help the compound maintain close contact with the active site, which is essential for inhibitory activity. The ligand remains well-positioned in the catalytic site, preventing substrate binding and thereby interfering with enzymatic function.

Alkyl interactions contribute to the stabilization of the ligand in hydrophobic regions of the enzyme. These interactions play a key role in maintaining the compound within the active site, reinforcing its inhibitory potential. By occupying crucial regions of the enzyme, the ligand can effectively hinder its activity, reducing the ability of the virus to process essential proteins for replication.

π-alkyl interactions further enhance ligand stability by facilitating strong stacking interactions with the enzyme. This contributes to a tighter and more effective binding, increasing the likelihood of successful inhibition. The presence of these interactions suggests that the ligand forms a robust complex with nsP2pro, which is critical for blocking its catalytic activity.

The combination of these interactions ensures that the ligand remains tightly bound to the enzyme, potentially preventing the virus from replicating efficiently. The strong binding affinity observed in molecular docking supports the hypothesis that these secondary metabolites could act as effective inhibitors. However, further molecular dynamics simulations are required to confirm the long-term stability of these interactions. Additionally, in vitro and in vivo studies are necessary to validate their efficacy as potential antiviral agents.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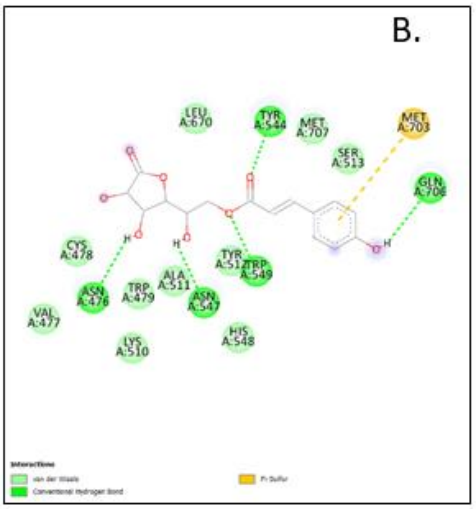
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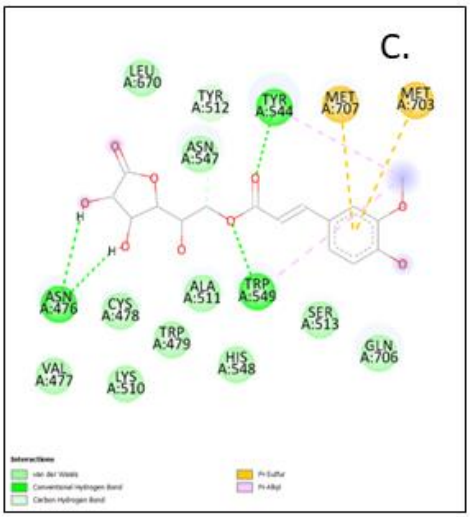
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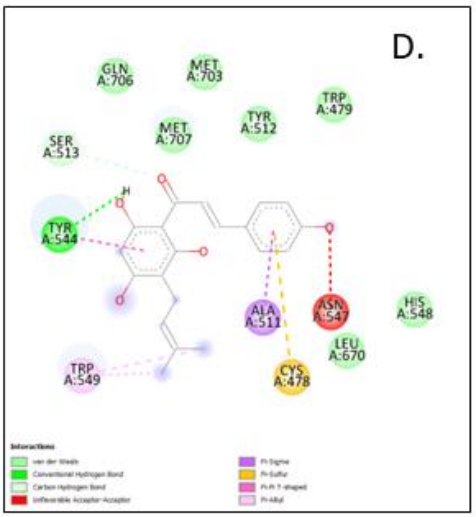
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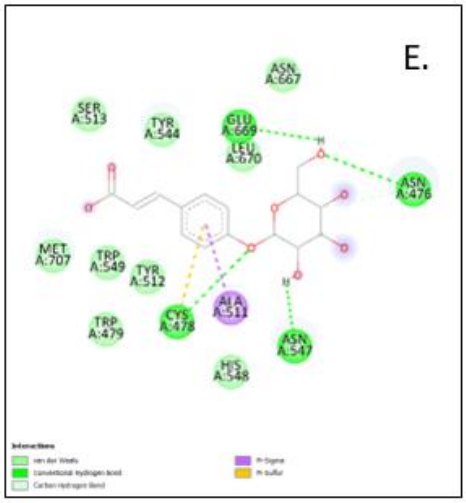
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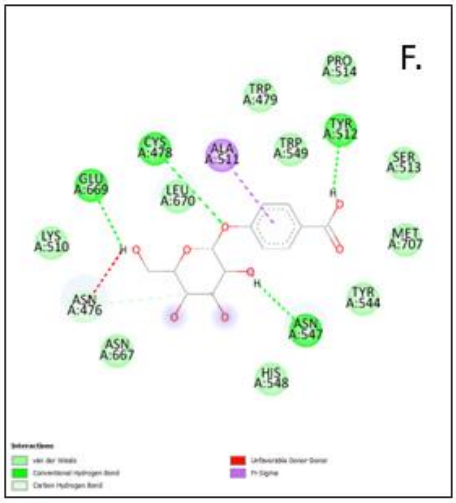
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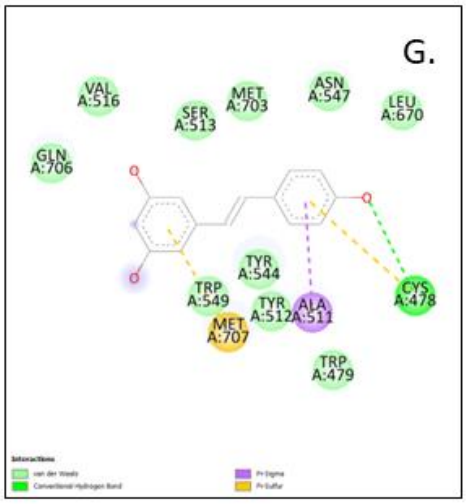
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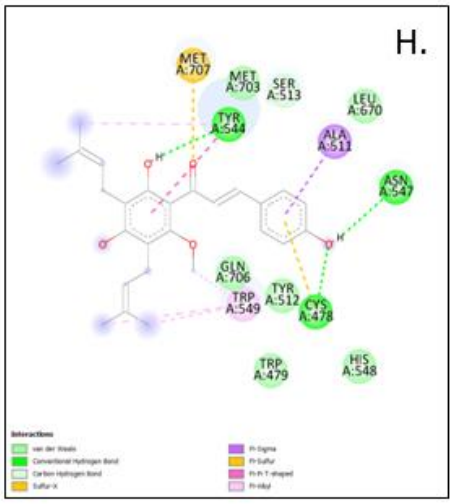
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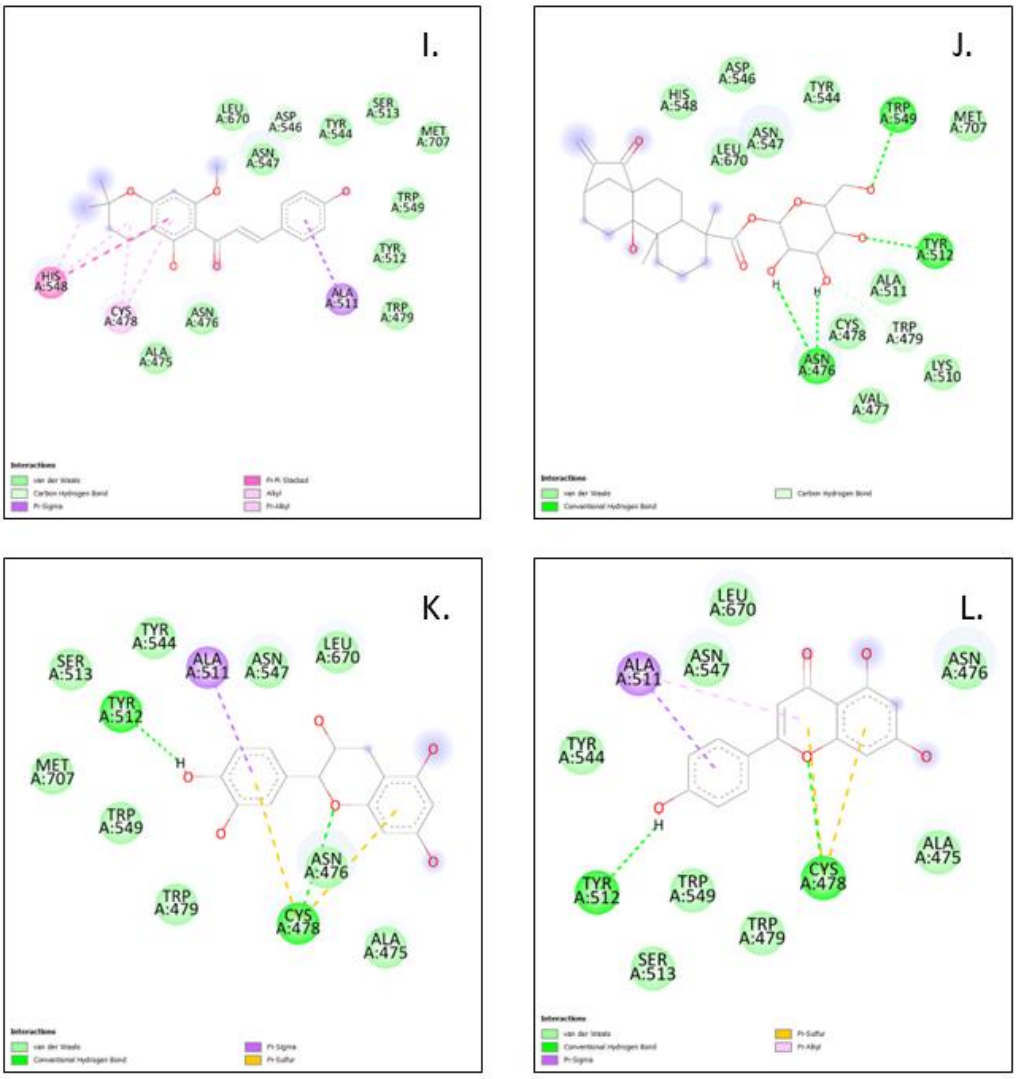

Figure 1: The 2D interactions

(A)3'-Geranylchalconaringenin (B)[(2S)-2-[(2R,3S,4S)-3,4-dihydroxy-5-oxooxolan-2-yl]-2-hydroxyethyl] 3-(4-hydroxyphenyl)prop-2-enoate (C)[(2S)-2-[(2R,3S,4S)-3,4-dihydroxy-5-oxooxolan-2-yl]-2-hydroxyethyl] 3-(4-hydroxy-3-methoxyphenyl)prop-2-enoate (D)Desmethylxanthohumol (E)(2Z)-3-(4-{[(2S,3R,4S,5S,6R)-3,4,5-trihydroxy-6-(hydroxymethyl)oxan-2-yl]oxy}phenyl)prop-2-enoic acid (F) 4-(beta-D-glucosyloxy)benzoic acid (G) Resveratrol (H) 5'-Prenylxanthohumol (I) 1-(5-Hydroxy-7-methoxy-2,2-dimethyl-3,4-dihydrochromen-6-yl)-3-(4-hydroxyphenyl)prop-2-en-1-one (J) ent-9-Hydroxy-15-oxokaur-16-en-19-oic acid beta-D-glucopyranosyl ester (K) Epicatechin (L) Apigenin

(A.) 3'-Geranylchalconaringenin showed the highest binding affinity (-7.2 kcal/mol), suggesting strong interactions with the nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Van der Waals, π-π T-shaped, alkyl, and π-alkyl interactions. One Conventional Hydrogen bond was established with amino acid Tyr544. Twelve Van der Waals interactions were established with amino acid residues Asn476, Trp479, Tyr512, Ser513, Pro514, Glu515, Val542, Asn547, Phe558, Glu669, Met703, and Gln706. These interactions help the ligand fit strongly into the nsP2pro active site, and they contribute to the overall binding stability of the ligand. Five alkyl and π-alkyl interactions were established with Cys478, Ala511, Lys556, Leu670, and Met707. Alkyl interactions occur between hydrophobic regions of the ligand and amino acid side chains, and π-alkyl interactions involve aromatic rings in the ligand interacting with hydrophobic residues these interactions enhance ligand stability within the hydrophobic pocket of the enzyme. These interactions suggest the compound has a strong affinity for the binding pocket. These interactions increase binding strength by stacking the aromatic rings in the ligand with amino acid side chains, and these interactions mimic the interactions of natural inhibitors of nsP2pro, suggesting the ligand may have inhibitory effects.

(B.) [(2S)-2-[(2R,3S,4S)-3,4-dihydroxy-5-oxooxolan-2-yl]-2-hydroxyethyl]3-(4-hydroxyphenyl)prop-2-enoate with the binding energy affinity -6.7 kcal/mol, suggesting strong interactions with the nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Van der Waals, and π-Sulphur interactions. Five Conventional Hydrogen bond interactions were established with amino acid residues Asn476, Tyr544, Asn547, Trp549, and Gln706. Ten van der Waals interactions were established with amino acid residues Val477, Cys478, Trp479, Lys510, Ala511, Tyr512, Ser513, His548, Leu670, and Met707.

(C.) [(2S)-2-[(2R,3S,4S)-3,4-dihydroxy-5-oxooxolan-2-yl]-2-hydroxyethyl]3-(4-hydroxy-3-methoxyphenyl)prop-2-enoate with the binding energy -6.6 suggesting strong interactions with nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Carbon-Hydrogen bonds, Van der Waals, π-Sulphur, alkyl and π-alkyl interactions. Three Conventional Hydrogen bonds were established with amino acid Asn476, Trp549, Tyr544. One hydrogen bond were established with amino acid Tyr512, Ten Van der Waals interactions were established

with amino acid residues Val477, Cys478, Trp479, Lys510, Ala511, Tyr512, Ser513, His548, Leu670, Met707, and two π-Sulphur interactions were established with amino acid residues Met703, Met707 and Two alkyl and π-alkyl interactions were established with Tyr544, Trp549.

(D.) Desmethylxanthohumol with binding energy affinity -6.5 suggesting strong interactions with nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Hydrogen bonds, Van der Waals, π-Sulphur bonds, π-σ bonds, alkyl and π-alkyl interactions, and unfavourable acceptor-acceptor interactions. One Conventional Hydrogen bond was established with amino acid residue Tyr544. One Hydrogen bond with amino acid Ser513. Seven Van der Waals interactions were established with amino acids residues Trp479, Tyr512, His548, Leu670, Met703, Gln706, Met707. One π-Sulphur bond with amino acid residue Cys478, one π-π T-shaped bond is established with amino acid residue Tyr544, one alkyl and π-alkyl interactions were established with amino acid Trp549, one π-σ bond with amino acid Ala511, and unfavourable acceptor-acceptor interaction was established with amino acid residue Asn547.

(E.) (2Z)-3-(4-{[(2S,3R,4S,5S,6R)-3,4,5-trihydroxy-6-(hydroxymethyl)oxan-2-yl]oxy}phenyl)prop-2-enoic acid with binding energy affinity -6.4 suggesting strong interactions with nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Hydrogen bonds, Van der Waals, π-Sulphur bonds, and π-σ bonds. Four Conventional Hydrogen bonds were established with amino acid residues Asn476, Cys478, Asn547, Glu669. One Hydrogen bond with amino acid Asn476. Nine Van der Waals interactions were established with amino acids residues Trp479, Tyr512, Ser513, Tyr544, His548, Trp549, Asn667, Leu670, Met707. One π-Sulphur bond with amino acid residue Cys478, and one π-σ bond with amino acid Ala511.

(F.) 4-(beta-D-glucosyloxy)benzoic acid with binding energy affinity -6.3 suggesting strong interactions with nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Hydrogen bonds, Van der Waals, π-σ bonds, and unfavourable donor interactions. Four Conventional Hydrogen bonds were established with amino acid residues Cys478, Tyr512, Asn547, Glu669. One Hydrogen bond with amino acid Asn476. Nine Van der Waals interactions were established with amino acids residues Trp479, Lys510, Ser513, Pro514, Tyr544, His548, Trp549,

Asn667, Met707. One π-σ bond with amino acid Ala511, and one unfavourable donor interactions with amino acid residue Asn476.

(G.) Resveratrol with binding energy affinity -6.3 suggesting strong interactions with nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Van der Waals, π-Sulphur bonds, and π-σ bonds. One Conventional Hydrogen bond was established with amino acid residues Cys478. Ten Van der Waals interactions were established with amino acids residues Trp479, Tyr512, Ser513, Val516, Tyr544, Asn547, Trp549, Leu670, Met703, Gln706. One π-Sulphur bond was established with amino acid residue Met707, and One π-σ bond with amino acid Ala511.

(H.) 5'-Prenylxanthohumol with binding energy affinity -6.3 suggesting strong interactions with nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Hydrogen bonds, Van der Waals, π-Sulphur bonds, Sulphur-X bonds, π-π T-shaped bonds, alkyl and π-alkyl interactions and π-σ bonds. Three Conventional Hydrogen bonds were established with amino acid residues Cys478, Tyr544, Asn547. One Hydrogen bond with amino acid Ser513. Six Van der Waals interactions were established with amino acids residues Trp479, Tyr512, His548, Leu670, Met703, Gln706. One π-Sulphur bond was established with amino acid residue Met707. One Sulphur-X bond was established with amino acid Met707. One π-π T-shaped bond was established with amino acid Tyr544. One alkyl and π-alkyl interactions were established with amino acid Trp549, and one π-σ bond with amino acid Ala511.

(I.) 1-(5-Hydroxy-7-methoxy-2,2-dimethyl-3,4-dihydrochromen-6-yl)-3-(4-hydroxyphenyl)prop-2-en-1-one with binding energy affinity -6.3 suggesting strong interactions with nsP2pro active site. The 2D interaction plot demonstrated the presence of Hydrogen bonds, Van der Waal, π-π Stacked bonds, alkyl and π-alkyl interactions and π-σ bonds. One Hydrogen bond with amino acid Asp546. Ten Van der Waals interactions were established with amino acids residues Ala475, Asn476, Trp479, Tyr512, Ser513, Tyr544, Asn547, Trp549, Leu670, Met707. One π-π Stacked bond was established with amino acid His548. Two alkyl and π-alkyl interactions were established with amino acid Cys478, His548, and one π-σ bond with amino acid Ala511.

(J.) ent-9-Hydroxy-15-oxokaur-16-en-19-oic acid beta-D-glucopyranosyl ester with binding energy affinity -6.3 suggesting strong interactions with nsP2pro active site. The

2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Hydrogen bonds, Van der Waals. Three Conventional Hydrogen bonds were established with amino acid residues Asn476, Tyr512, Trp549. One Hydrogen bond with amino acid Trp479. Ten Van der Waals interactions were established with amino acids residues Val477, Cys478, Lys510, Ala511, Tyr512, Tyr544, Asp546, Asn547, His548, Met707.

(K.) Epicatechin with binding energy affinity -6.2 suggesting strong interactions with nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Van der Waal, π-Sulphur bonds, and π-σ bonds. Two Conventional Hydrogen bonds were established with amino acid residues Cys478, Tyr512. Nine Van der Waals interactions were established with amino acids residues Ala475, Asn476, Trp479, Ser513, Tyr544, Asn547, Trp549, Leu670, Met707. One π-Sulphur bond was established with amino acid Cys478, and one π-σ bond with amino acid Ala511.

(L.) Apigenin with binding energy affinity -6.2 suggesting strong interactions with nsP2pro active site. The 2D interaction plot demonstrated the presence of Conventional Hydrogen bonds, Van der Waal, π-Sulphur bonds, alkyl and π-alkyl interactions, and π-σ bonds. Two Conventional Hydrogen bonds were established with amino acid residues Cys478 and Tyr512. Eight Van der Waals interactions were established with amino acids residues Ala475, Asn476, Trp479, Ser513, Tyr544, Asn547, Trp549, and Leu670. One π-Sulphur bond was established with amino acid Cys478, one alkyl and π-alkyl bond was established with amino acid residue Ala511 and one π-σ bond with amino acid Ala511.

6. CONCLUSION

In this study, we utilized a range of in-silico techniques to explore the binding interactions of potential inhibitors with the active site of the 4ZTB enzyme, which is associated with species commonly found in the Polypodiaceae family. Notably, the binding of the test compounds did not result in any conformational alterations in the enzyme's structure. The integration of computational approaches, including molecular docking, pharmacophore modeling, and virtual screening, demonstrates a cost-effective and time-efficient strategy for identifying promising therapeutic candidates for treating Chikungunya, thereby accelerating the early stages of drug discovery.

7. FUTURE DIRECTIONS

Future research on Chikungunya virus nsP2 protease (nsP2pro) should focus on increasing our understanding of its interaction dynamics with bioactive secondary metabolites derived from the *Polypodiaceae* family. High-throughput virtual screening combined with advanced molecular dynamics simulations can further elucidate the stability and specificity of these interactions at the atomic level. Experimental validation through in vitro enzymatic assays and cell-based antiviral studies will be crucial to confirm computational findings and assess cytotoxicity and efficacy. Ultimately, this integrated approach may lead to the development of novel, plant-based therapeutic agents targeting nsP2pro, offering a promising antiviral strategy against Chikungunya virus infections.

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